

Silicification, not oxygenation: a compiled-record test of the redox-preservation hypothesis for the Precambrian microfossil record

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Abstract

Low atmospheric oxygen before the Great Oxidation Event (GOE, ~ 2.4 Ga) should, in principle, have slowed oxidative degradation of organic matter and favoured fossil preservation; yet morphologically diagnostic microfossils are comparatively sparse before the GOE and diversify afterwards. We tested the attractive interpretation that this pattern reflects a redox-driven shift in preservation, by compiling published preservation, redox-proxy and silica-cycle data across the GOE for six chert-hosted biotas (Strelley Pool, Moodies and Tumbiana, pre-GOE; Belcher, Gunflint and Sokoman, post-GOE) and a cross-GOE shale redox compilation. Three findings reshape the question. (1) Morphological fidelity tracks *early silicification*, not the GOE: the pre-GOE Strelley Pool biota is silica-encapsulated and morphology-rich, falsifying any simple pre/post-GOE contrast. (2) The premise that the redox regime shifted cleanly across the GOE is not supported: ferruginous conditions dominate on both sides in the global iron-speciation compilation. (3) Seawater silica was supersaturated throughout the Precambrian and was not drawn down until the Neoproterozoic–Phanerozoic radiation of biosilicifiers, so the silica cycle did not shift across the GOE either. Together these reject the specific mechanisms proposed for a redox-driven preservation shift (sulfate-enabled pyritisation; a GOE silica-cycle lever) and implicate silicification—a facies/diagenetic variable largely decoupled from marine redox—as the dominant control. We conclude that the apparent post-GOE “rise” of microfossils is primarily a facies/sampling signal rather than an oxygenation signal, and we outline the narrower conditions under which the GOE could still exert a second-order taphonomic influence.

Keywords: Great Oxidation Event; taphonomy; silicification; Precambrian microfossils; iron speciation; silica cycle; kerogen; Raman spectroscopy.

1 Introduction

Before the GOE, atmospheric O_2 was below $\sim 10^{-5}$ PAL — low enough that mass-independent sulfur isotope fractionation (S-MIF) was continuously expressed (Pavlov & Kasting, 2002; Farquhar et al., 2000) — and oceans were anoxic and ferruginous with seawater sulfate $\lesssim 200$ μM (Habicht et al., 2002; Crowe et al., 2014). Under such conditions aerobic respiration, the most efficient organic-matter degradation pathway, was suppressed. It is therefore tempting to expect the Archean to have been a high-fidelity archive of early life, and equally tempting to read the post-GOE diversification of the microfossil record (Javaux et al., 2018; Wacey et al., 2014) as a preservational—or biological—consequence of oxygenation. The two readings are not exclusive, and distinguishing them is the deeper problem this paper addresses.

We began from a working hypothesis: that preservation of organic carbon and of morphology are decoupled, and that the GOE-driven sulfate rise made microbial pyritisation newly available, so that post-GOE morphology reflected a redox-enabled preservation pathway rather than biomass. This report records what a compiled-record test of that hypothesis revealed, and how the hypothesis was revised as a result. We treat the evolution of the interpretation as part of the scientific record rather than disguising it: a hypothesis that survives empirical stress is worth more than one never tested.

2 Background

2.1 The GOE and the redox landscape

The GOE is pinned to ~ 2.32 – 2.45 Ga by the disappearance of S-MIF (Bekker et al., 2004; Farquhar & Wing, 2003), but was prolonged and oscillatory (Lyons et al., 2014; Garduño Ruiz et al., 2024), with proposed drivers including organic-carbon burial during the Lomagundi–Jatuli excursion (Bekker & Holland, 2012; Hodgskiss et al., 2023), hydrogen escape (Catling et al., 2001) and a shift to subaerial volcanism (Kump & Barley, 2007). Mid-Proterozoic O_2 was low (~ 1 – 10% PAL or $\leq 0.1\%$ PAL; Lyons et al., 2014; Planavsky et al., 2014) and the deep ocean remained anoxic (Canfield, 1998). Local redox is reconstructed from iron speciation ($Fe_{HR}/Fe_T > 0.38$; Poulton & Canfield, 2005, 2011), chromium isotopes (Frei et al., 2009) and Mo–U–N systematics (Scott et al., 2008; Garvin et al., 2009). Seawater sulfate rose across the GOE from μM to mM levels (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014).

2.2 The Precambrian microfossil record

Archean microfossils are sparse and contested: the ~ 3.46 Ga Apex chert structures (Schopf, 1993) were reinterpreted as abiogenic artefacts (Brasier et al., 2002), and many Archean biotas are mat-related and morphologically simple. A widely noted pattern is that morphologically diagnostic, taxonomically resolvable assemblages become abundant only after the GOE, peaking in the Gunflint-type biotas at ~ 1.9 Ga (Javaux et al., 2018; Wacey et al., 2014).

2.3 Two preservation modes, and the mineral templating race

Organic-carbon survival is governed by oxygen-exposure time, sedimentation rate, recalcitrance, and mineral (iron-) surface protection (Canfield, 1994; Hedges & Keil, 1995; Hartnett et al., 1998; Lalonde et al., 2012). Morphological fidelity, in contrast, is won or lost in a race between decay and early-diagenetic mineral templating: cellular structure is preserved only when silica, pyrite, phosphate or clay minerals impregnate or encrust cells *before* decay collapses morphology (Wacey et al., 2012). The two modes share decay as an antagonist but depend on different, independently variable mineral pathways.

2.4 Raman spectroscopy and the maturity control

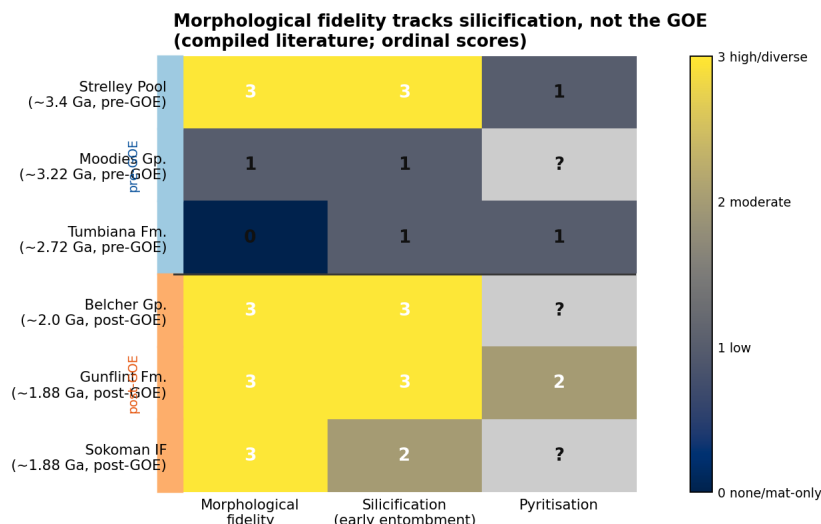
Raman spectroscopy of carbonaceous material (CM) provides both a peak-maturity geothermometer (Beyssac et al., 2002) (via $R_2 = A_{D1}/(A_{D1} + A_{G1} + A_{G2})$) and a measure of CM structural order aromaticity (Qu et al., 2019), allowing samples to be compared for thermal history and for the structural imprint of differing preservation pathways.

2.5 Silicification and the Precambrian silica cycle

Silica entombment preserves labile molecular signatures (Alleon et al., 2016) and underpins exceptional preservation from the Archean to the Ediacaran (Tarhan et al., 2016). Critically, before the evolution of biosilicifiers, Precambrian seawater was supersaturated with respect to amorphous silica (~ 2 mM Si; ≥ 60 ppm SiO_2 , versus ~ 0.1 ppm today), with abiological precipitation the dominant sink (Siever, 1992; Perry & Lefticariu, 2003). The drawdown of dissolved silica is a Neoproterozoic–Phanerozoic story — driven by sponges, then radiolaria (Ordovician) and diatoms (Mesozoic–Cenozoic) — *not* a GOE story (Table 1).

3 Approach: a compiled-record test

We compiled published preservation, redox and silica data across the GOE into two panels (data available with the study). **Panel A:** six chert-hosted biotas spanning the GOE, scored ordinally (0–3) for morphological fidelity, early silicification, and pyritisation (sources in Fig. 1). **Panel B:** a



Scores compiled from Delarue 2020/2021, Sugitani 2010, Lepot 2009/2017, Alleen 2016, Wacey 2013, Homann 2015, Hodgskiss & Sperling 2020. Hypothesis-generating, not statistical. '?' = not found in open literature.

Figure 1: Compiled literature scores (0–3; grey = “?” not found) for six chert-hosted biotas. Morphological fidelity and silicification columns share the same pattern; the GOE side does not. Scores compiled from (Delarue et al., 2020, 2021; Sugitani et al., 2010; Lepot et al., 2009; Decreane et al., 2021; Lepot et al., 2017; Alleen et al., 2016a; Wacey et al., 2013; Homann et al., 2015; Hodgskiss & Sperling, 2020). Ordinal and hypothesis-generating, not statistical.

cross-GOE shale redox compilation ($\text{Fe}_{\text{HR}}/\text{Fe}_{\text{T}}$, $\text{Fe}_{\text{py}}/\text{Fe}_{\text{HR}}$, sulfate/trace-metal proxies). We treat the scores as hypothesis-generating, not statistical, and report gaps honestly. All entries are sourced; no values were estimated.

4 Findings

4.1 Morphological fidelity tracks silicification, not the GOE

Across all six biotas, morphological fidelity co-varies with early silica encapsulation and is uncorrelated with GOE side (Fig. 1). The decisive counter-example to a GOE/redox driver is **Strelley Pool** (~3.4 Ga, ~1 Gyr before the GOE): silica-encapsulated, Raman ~300 °C, and morphology-rich (lenticular forms, chains/clusters, and “tail-like” microfossils; Delarue et al., 2020, 2021; Sugitani et al., 2010). If redox/GOE drove morphological richness, Strelley Pool should be poor; it is not. Conversely, the pre-GOE Tumbiana and Moodies biotas are morphology-poor precisely where silica entombment is weak or absent (carbonate-hosted globules; siliciclastic mats; Lepot et al., 2009; Homann et al., 2015; Decreane et al., 2021). The post-GOE Gunflint, Sokoman and Belcher biotas are silica-encapsulated and morphology-rich (Lepot et al., 2017; Hodgskiss & Sperling, 2020). Where preservation is resolved at the nanoscale (Gunflint), the preservative pathway is explicitly early silicification plus Fe-biomineralization (Lepot et al., 2017; Alleen et al., 2016a).

4.2 Pyritisation is destructive, not preservative, in the best-preserved biotas

In Gunflint, pyritisation occurs in scarce patches, in a locally sulfate-limited setting, and “nearly obliterates organic structures” (Wacey et al., 2013; Lepot et al., 2017); the preserved morphologies carry intra-cellular greenalite + siderite, not pyrite (Lepot et al., 2017). Pyritisation also occurred pre-GOE (Tumbiana micropyrates and bacterial-sulfate-reduction-driven sulfurization; Decreane et al., 2021; Lepot et al., 2009). The original sulfate→pyritisation mechanism is therefore rejected on three independent grounds: pyritisation is pre-GOE, locally sulfate-limited even post-GOE, and destructive where it occurs.

Table 1: Secular state of the marine silica cycle. The GOE (2.4 Ga) sits entirely within the high-silica Precambrian regime; the drawdown is a much later, biologically driven event. After (Siever, 1992; Perry & Leticariu, 2003; Earth-Science Reviews, 2023).

Interval	Seawater DSi	Dominant Si sink	GOE relevance
Archean–Proterozoic	supersaturated (~ 2 mM)	abiological (chert, Fe-oxide/clay adsorption)	high on both sides
Neoproterozoic–early Palaeozoic	beginning to decline	early sponges, first radiolaria	post-dates GOE
Palaeozoic–Mesozoic	substantially drawn down	radiolaria, sponges; diatoms rise	post-dates GOE
Cenozoic–modern	~ 0.1 ppm	diatoms ($\sim 75\%$ of biogenic Si burial)	—

4.3 The redox regime did not shift cleanly across the GOE

The cross-GOE shale record does not show an uncontested redox shift. Ferruginous conditions dominate on both sides of the GOE, euxinia is spatially variable (e.g. Barney Creek photic euxinia, Sperling et al., 2014; Mt McRae transient “whiffs”, Anbar et al., 2007), and the global compilation shows ferruginous conditions dominating on both sides (no clean anoxic-to-oxic transition at the GOE; Poulton & Canfield, 2011; Sperling et al., 2014). The redox shift cannot therefore be assumed as the independent variable; it is itself a hypothesis, and one that diagenetic overprinting can further obscure.

4.4 The silica cycle did not shift across the GOE either

Because seawater silica was high throughout the Precambrian, the GOE could not have “switched on” silicification capacity. Any revised mechanism invoking a GOE silica-cycle lever therefore also fails: silicification *capacity* was available before and after the GOE alike.

5 Revised interpretation

The compiled-record test rejects both candidate GOE mechanisms — sulfate-enabled pyritisation (§4.2) and a GOE silica-cycle switch (§4.4) — while supporting the more general claim that morphological fidelity is gated by early silica encapsulation (§4.1). The inescapable conclusion is that the GOE is *not* the primary control on microfossil morphological preservation. Because silica entombment capacity was high on both sides of the GOE, the variation in fidelity across the record is governed by *which facies received early silicification* (a local depositional and early-diagenetic control: peritidal concentration, ferruginous silica precipitation, microbial silica nucleation), not by oxygenation. The apparent post-GOE “rise” of morphological microfossils is therefore most parsimoniously read as a facies and rock-volume/sampling signal, amplified where silica-encapsulating shallow-marine cherts are well-exposed and well-sampled (Gunflint, Belcher), rather than as a biological or redox signal.

We do not claim the GOE is taphonomically irrelevant. Two second-order channels remain plausible. First, the GOE changed the *decay community* (more O_2 , more sulfate post-GOE), altering the rate and selectivity of organic-matter degradation; but where silicification outpaced decay in silica-rich facies — as at pre-GOE Strelley Pool — this difference would not surface as a morphology contrast. Second, sulfate and the OM-sulfurization balance affect *what* organic matter survives (the S-poor/S-rich OM pools of Tumbiana, Lepot et al., 2009), influencing bulk kerogen character more than cellular fidelity. Both are quantifiable but neither is the dominant signal in the present compilation.

6 Implications

Reading the fossil record. Treating microfossil abundance or diversity as a direct proxy for biomass or biological innovation across the GOE is unreliable without accounting for the silicification/facies control. Quantitative estimates of Proterozoic diversity should be facies-corrected, not redox-corrected alone.

Unmasking versus evolution. The richness of OM in pre-GOE units without diagnostic morphology (Tumbiana globules; Moodies mats) is consistent with *latent, morphologically invisible* diversity; resolving whether post-GOE morphological novelty reflects evolution or unmasking still requires independent temporal evidence (molecular clocks, biomarkers), as recently emphasised in critiques of framework papers of this type.

Biosignatures on Mars and early Earth. The absence of morphological microfossils in an anoxic, poorly silicified succession does not entail the absence of life; the requisite variable is a silica-encapsulating facies, not oxygen.

7 Conclusions

A compiled-record test across the GOE falsifies the redox-preservation hypothesis in its specific forms: morphological fidelity tracks early silicification, not the GOE (pre-GOE Strelley Pool is morphology-rich); pyritisation is pre-GOE, locally sulfate-limited, and destructive; the bulk-redox shift across the GOE is itself contested; and the silica cycle did not shift across the GOE. The post-GOE “rise” of microfossils is best read as a facies/sampling signal rather than an oxygenation signal. The work illustrates the value of subjecting an attractive taphonomic narrative to an empirical, fiction-free compiled-record test before drawing biological or redox conclusions from first appearances. The narrower question of how the GOE reshaped decay and OM-sulfurisation remains open and is the appropriate focus of targeted new measurement.

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